Kernix 0.11.1 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

46 workloads

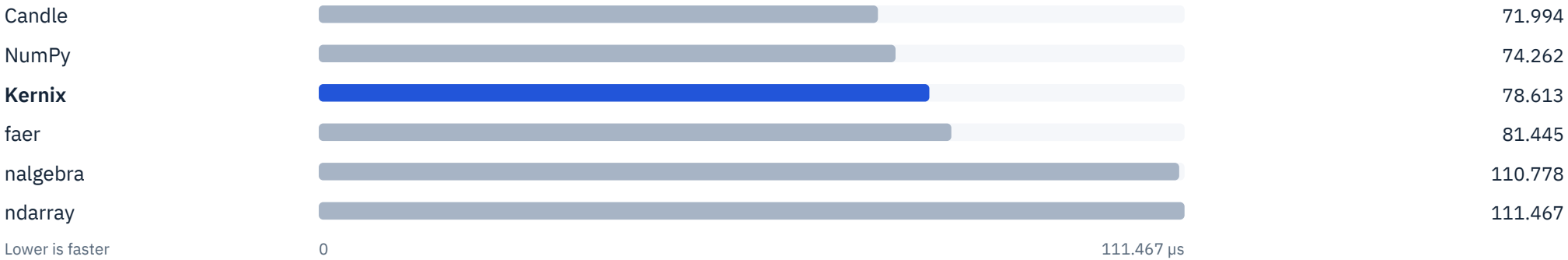
149 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

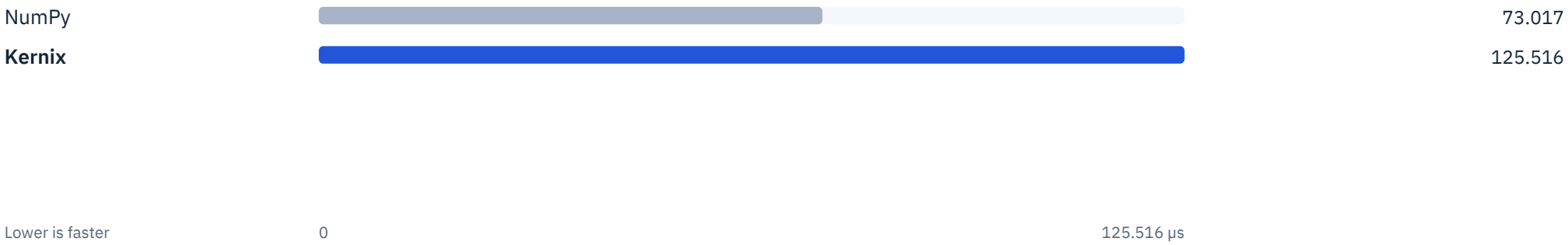
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



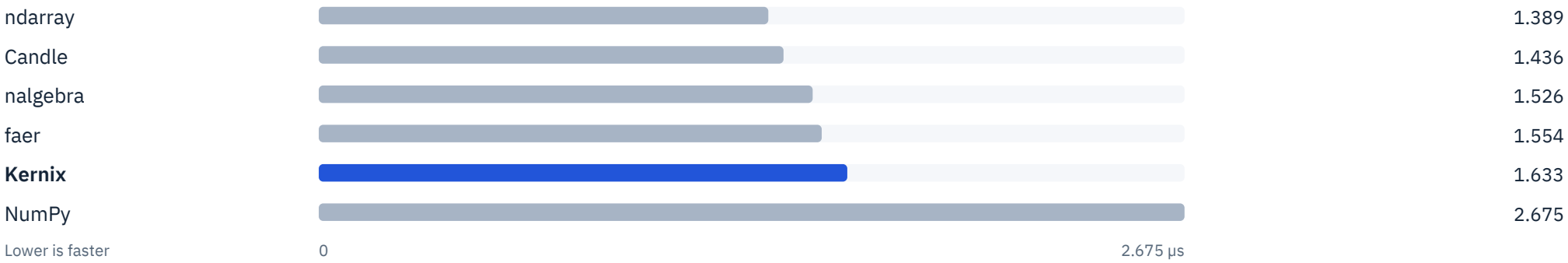
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

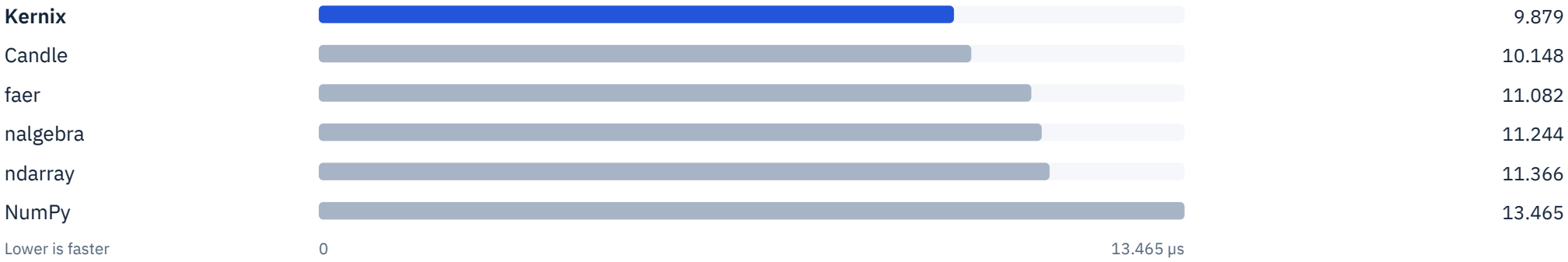
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



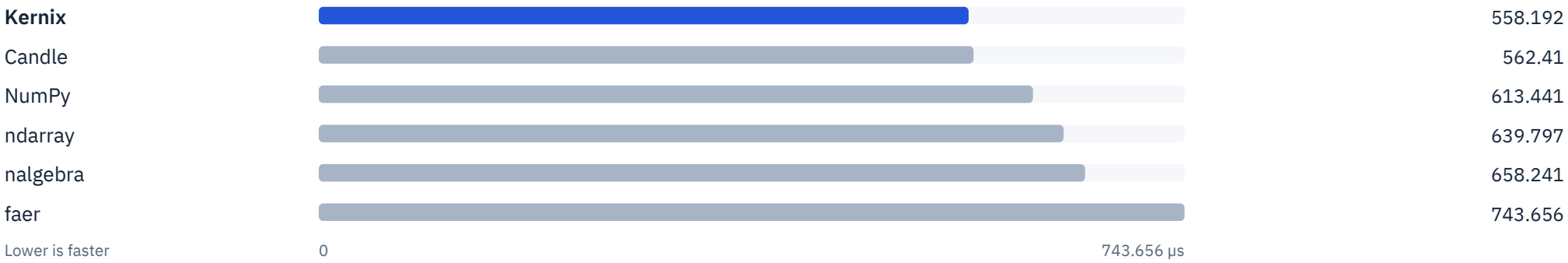
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



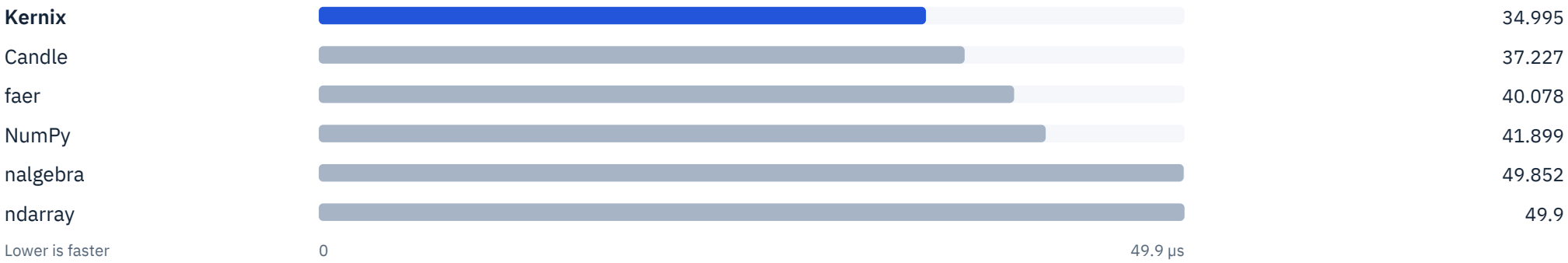
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



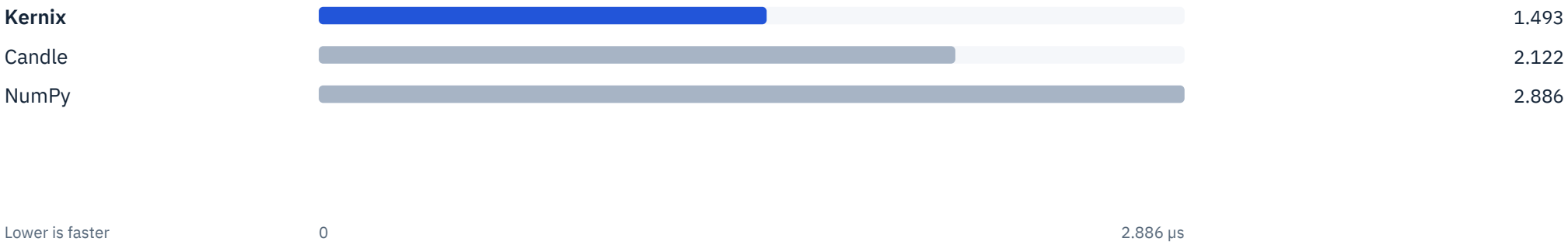
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

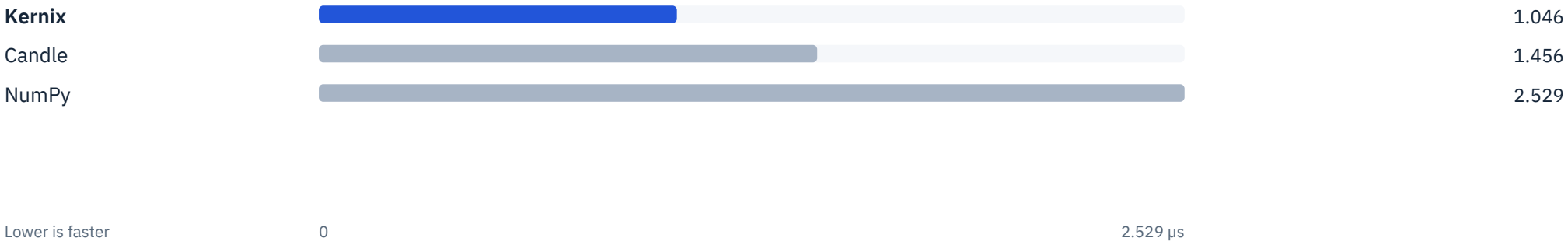
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

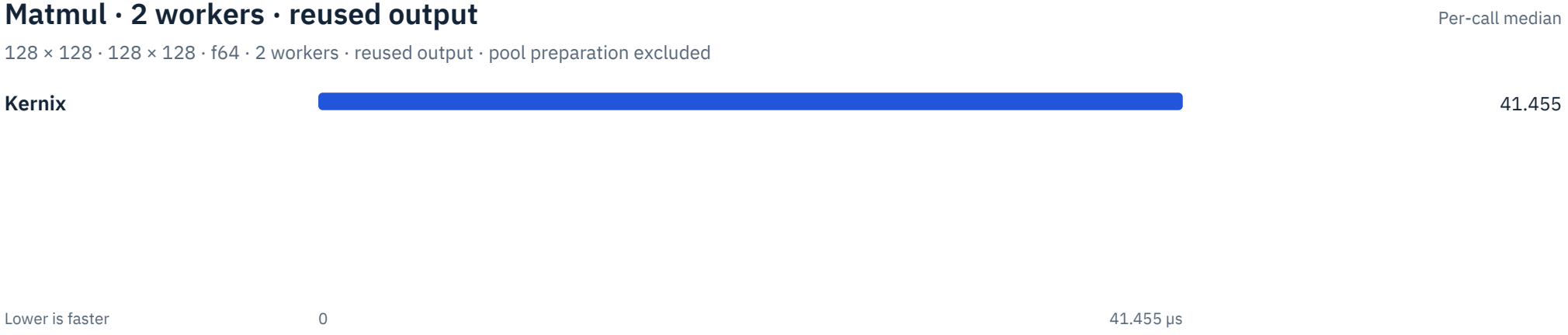
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded

Per-call median



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded

Per-call median



Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

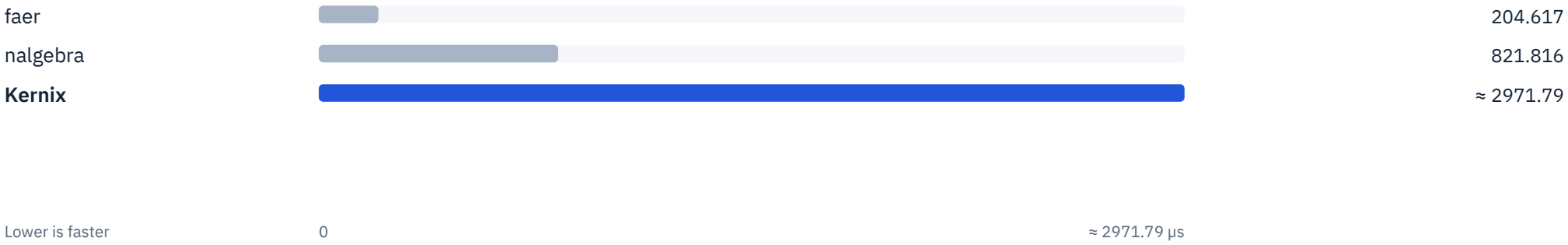
128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



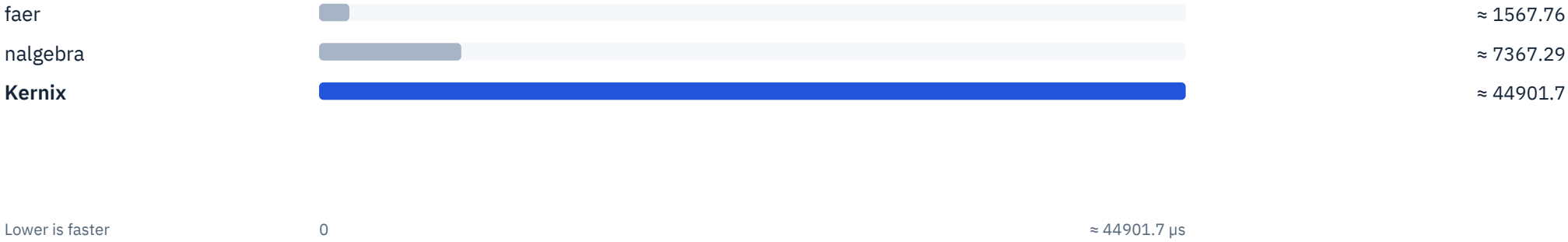
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

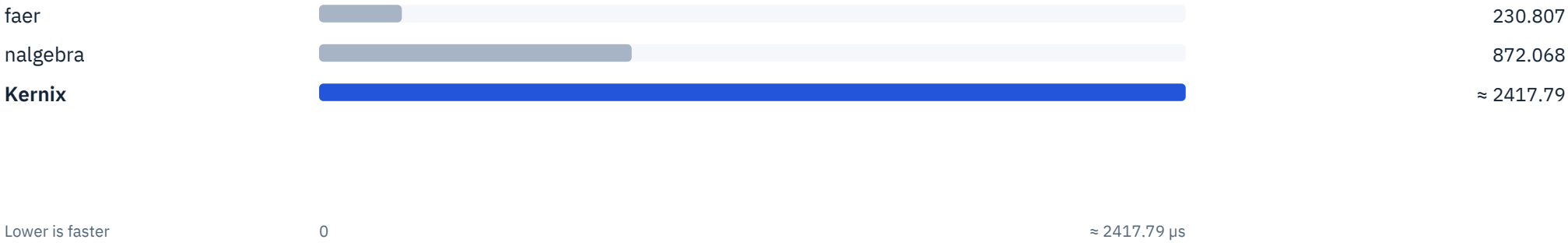
512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



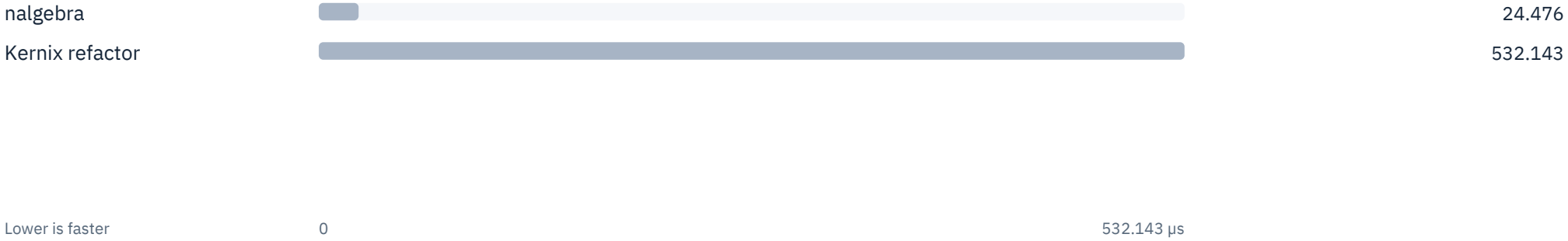
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

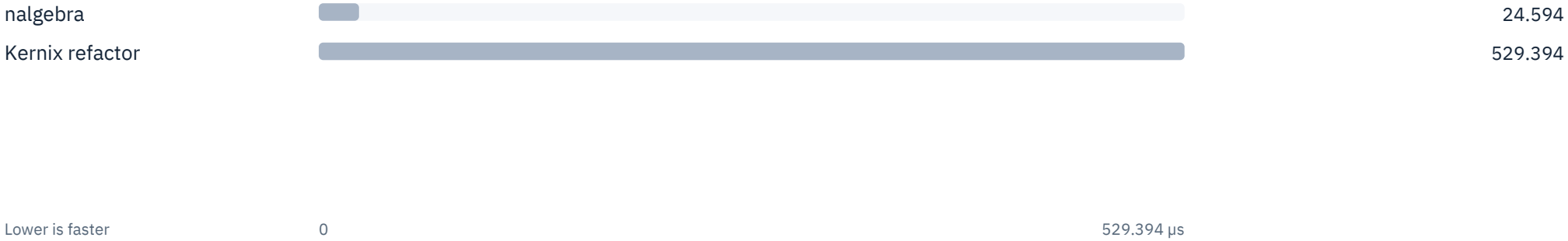
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

Per-call median

512 × 128 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

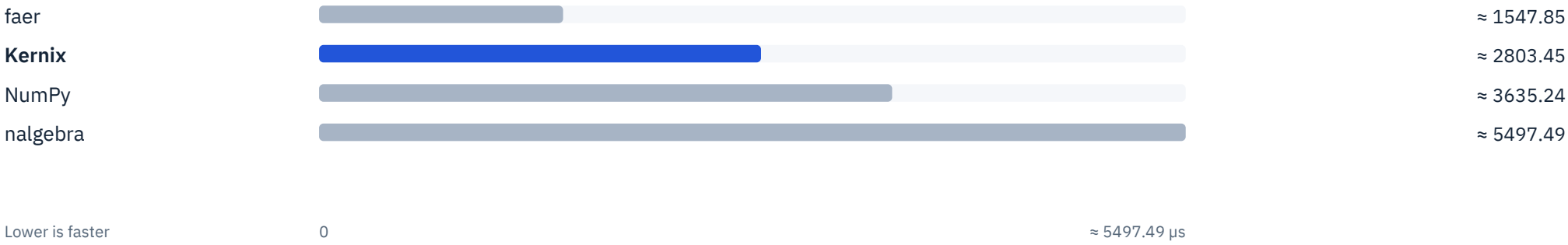
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



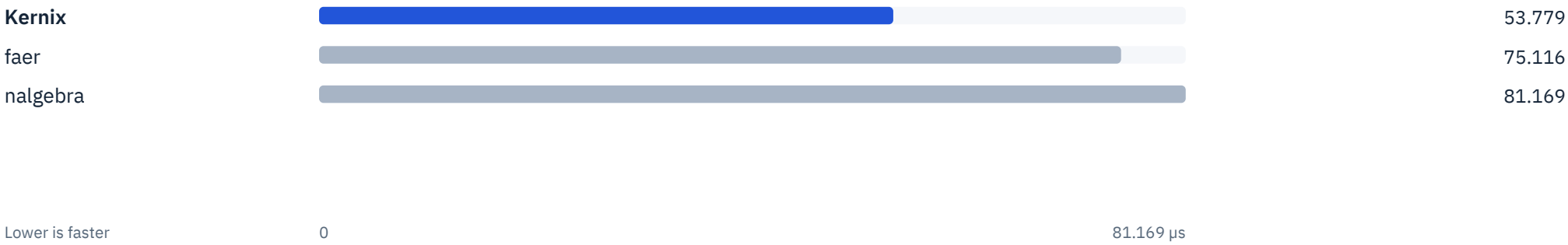
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

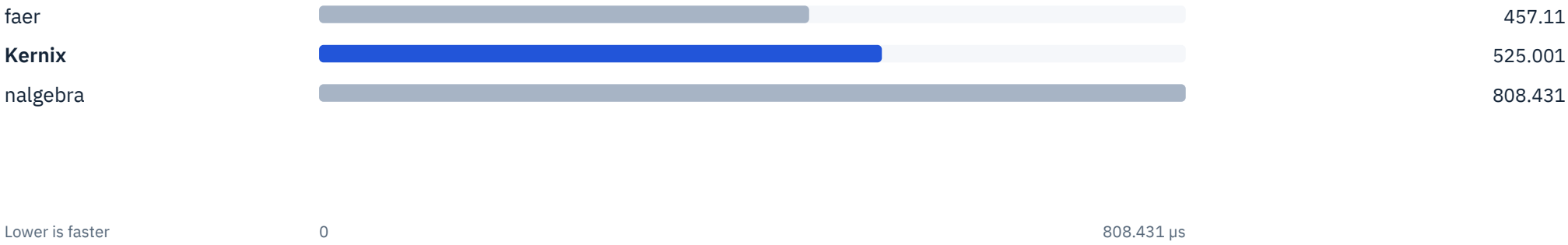
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



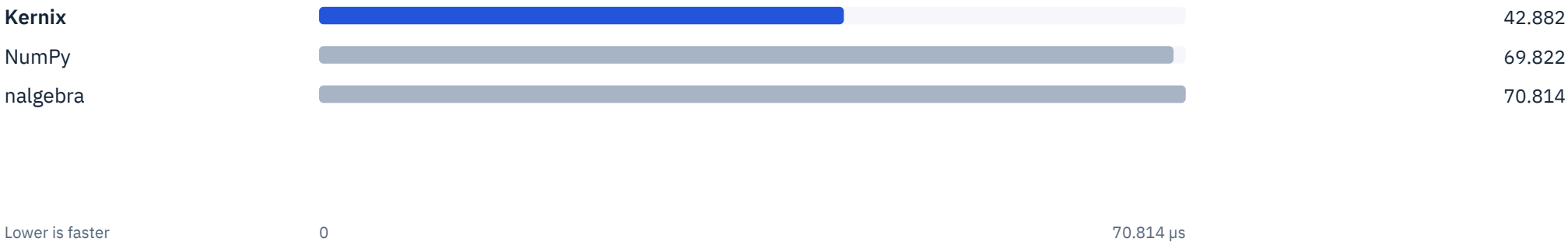
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

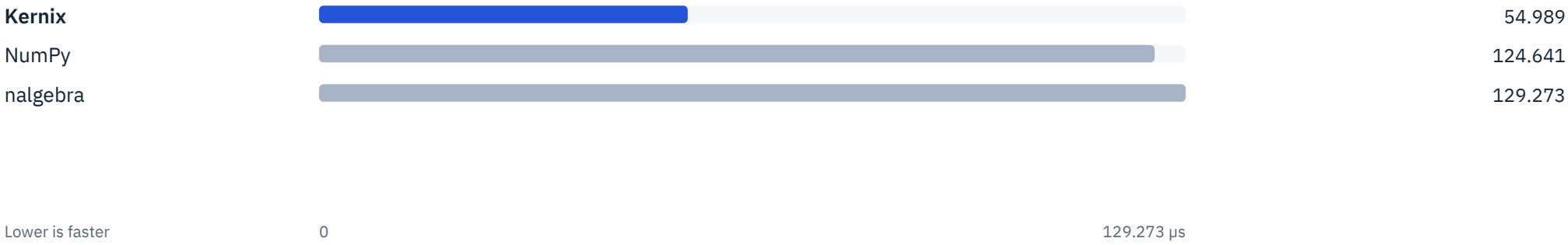
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



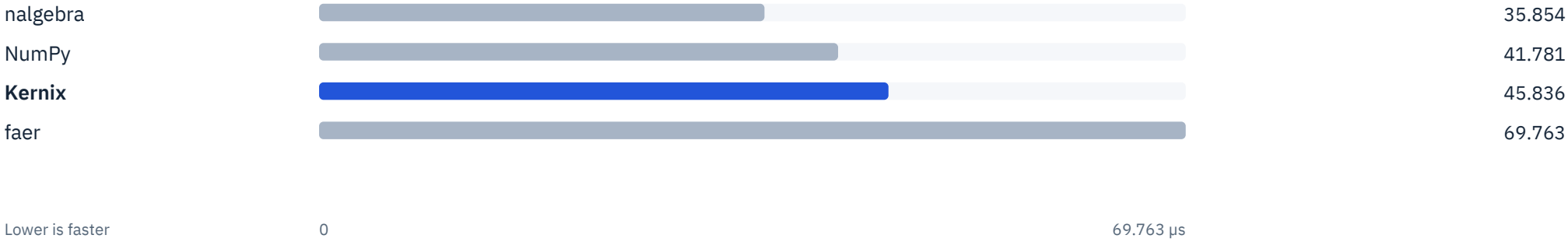
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



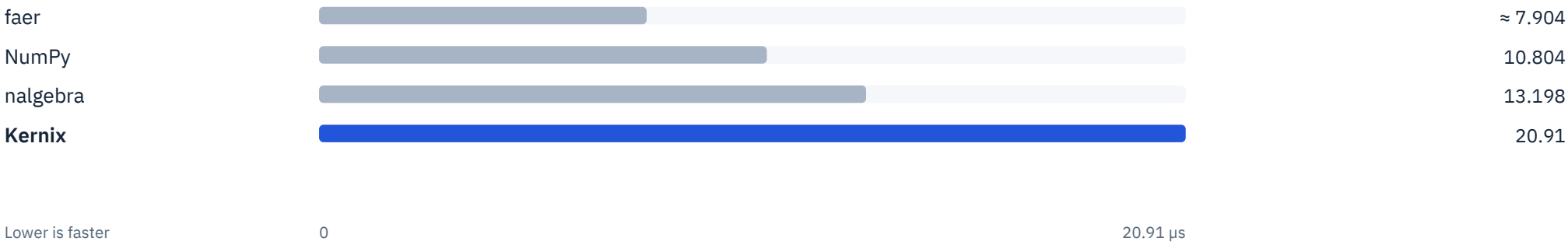
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



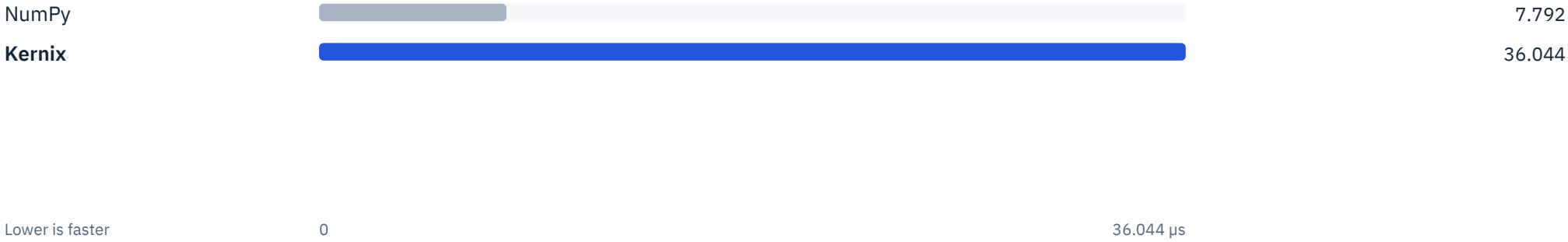
Kernix 0.11.1 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median

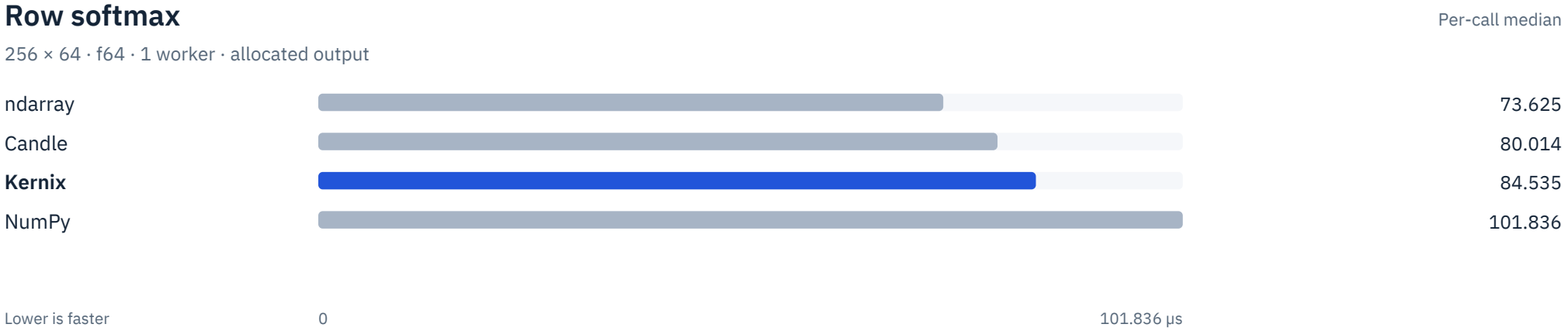


Kernix 0.11.1 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



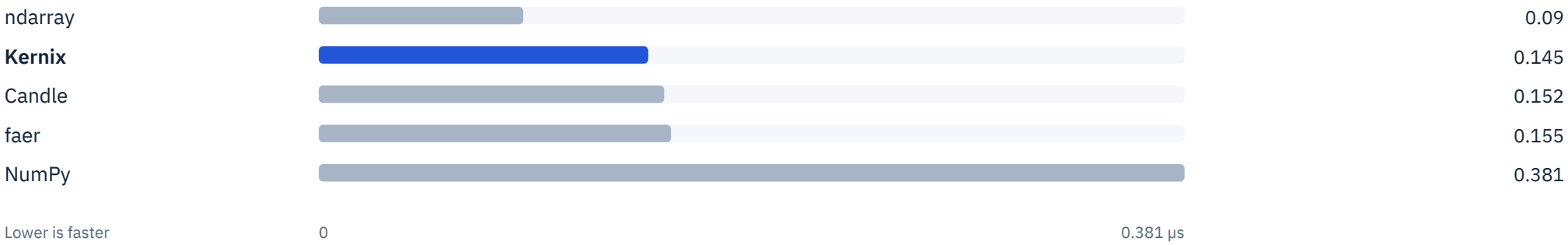
Kernix 0.11.1 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	73017	73.017
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	125516	125.516
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	36199	36.199
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	48376	48.376
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	74171	74.171
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	72806	72.806
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	37823	37.823
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	36309	36.309
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1526	1.526
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1633	1.633

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1389	1.389
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1436	1.436
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2675	2.675
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1554	1.554
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	13465	13.465
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11366	11.366
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	10148	10.148
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11244	11.244
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9879	9.879
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	11082	11.082

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	78613	78.613
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	74262	74.262
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	110778	110.778
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	71994	71.994
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	81445	81.445
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	111467	111.467
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	613441	613.441
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	658241	658.241
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	558192	558.192
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	562410	562.41

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	639797	639.797
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	743656	743.656
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5156	5.156
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6088	6.088
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6093	6.093
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5463	5.463
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	5713	5.713
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4689	4.689
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	41899	41.899
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	49900	49.9

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	49852	49.852
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	37227	37.227
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	34995	34.995
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	40078	40.078
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	411441	411.441
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	289080	289.08
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	311138	311.138
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	272518	272.518
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	273336	273.336
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	414792	414.792

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2886	2.886
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1493	1.493
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2122	2.122
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2529	2.529
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1046	1.046
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	1456	1.456
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	10988	10.988
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	41455	41.455
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	303263	303.263
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	23390	23.39

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	3919	3.919
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	3966	3.966
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	11728	11.728
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	2971787	2971.787
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	821816	821.816
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	204617	204.617
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	44901669	44901.669
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	1567762	1567.762
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	7367292	7367.292
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	872068	872.068

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	2417794	2417.794
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	230807	230.807
Cholesky rank-one update / nalgebra 256 × 256 · f64	24476	24.476
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	532143	532.143
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	529394	529.394
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	24594	24.594
Cholesky rank-one update / nalgebra 512 × 512 · f64	145180	145.18
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	2943494	2943.494
Thin QR / nalgebra 512 × 128 · f64	3123168	3123.168
Thin QR / NumPy 512 × 128 · f64	2492424	2492.424

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 512 × 128 · f64	1998504.9999999998	1998.5049999999998
Thin QR / Kernix 512 × 128 · f64	3593402	3593.402
Thin QR / Kernix 1024 × 256 · f64	29615796	29615.796
Thin QR / faer 1024 × 256 · f64	13279983	13279.983
Thin QR / NumPy 1024 × 256 · f64	18024881	18024.881
Thin QR / nalgebra 1024 × 256 · f64	29215404	29215.404
Cholesky factorization / nalgebra 512 × 512 · f64	5497489	5497.489
Cholesky factorization / faer 512 × 512 · f64	1547853	1547.853
Cholesky factorization / Kernix 512 × 512 · f64	2803451	2803.451
Cholesky factorization / NumPy 512 × 512 · f64	3635237	3635.237

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 96 × 32 · f64	81169	81.169
Column-pivoted QR / faer 96 × 32 · f64	75116	75.116
Column-pivoted QR / Kernix 96 × 32 · f64	53779	53.779
Column-pivoted QR / Kernix 256 × 64 · f64	525001	525.001
Column-pivoted QR / faer 256 × 64 · f64	457110	457.11
Column-pivoted QR / nalgebra 256 × 64 · f64	808431	808.431
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	42882	42.882
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	70814	70.814
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	69822	69.822
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	129273	129.273

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	124641	124.641
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	54989	54.989
Thin QR / faer 96 × 32 · f64	69763	69.763
Thin QR / NumPy 96 × 32 · f64	41781	41.781
Thin QR / nalgebra 96 × 32 · f64	35854	35.854
Thin QR / Kernix 96 × 32 · f64	45836	45.836
Thin QR / nalgebra 256 × 64 · f64	372020	372.02
Thin QR / Kernix 256 × 64 · f64	477593	477.593
Thin QR / faer 256 × 64 · f64	427141	427.141
Thin QR / NumPy 256 × 64 · f64	365029	365.029

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 64 × 64 · f64	10804	10.804
Cholesky factorization / Kernix 64 × 64 · f64	20910	20.91
Cholesky factorization / nalgebra 64 × 64 · f64	13198	13.198
Cholesky factorization / faer 64 × 64 · f64	7903.999999999999	7.903999999999999
Cholesky factorization / NumPy 128 × 128 · f64	76363	76.363
Cholesky factorization / faer 128 × 128 · f64	41318	41.318
Cholesky factorization / Kernix 128 × 128 · f64	110599	110.599
Cholesky factorization / nalgebra 128 × 128 · f64	101281	101.281
Cholesky factorization / nalgebra 256 × 256 · f64	673671	673.671
Cholesky factorization / faer 256 × 256 · f64	197118	197.118

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	529307	529.307
Cholesky factorization / Kernix 256 × 256 · f64	512362.00000000006	512.36200000000006
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3892	3.892
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6713	6.713
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4022	4.022
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2394	2.394
Model inference / Kernix batched normalized/sliced MLP · f64	2077	2.077
Model inference / Candle batched normalized/sliced MLP · f64	3245	3.245
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2605	2.605
Model recording + forward / Candle batched normalized/sliced MLP · f64	5127	5.127

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	3162	3.162
Model backward / Candle prepared graph → fresh gradients · f64	13517	13.517
Parameter update / Kernix transactional SGD, fixed gradients · f64	526	0.526
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	7792	7.792
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	36044	36.044
Row softmax / ndarray 256 × 64 · f64	73625	73.625
Row softmax / Candle 256 × 64 · f64	80014	80.014
Row softmax / Kernix 256 × 64 · f64	84535	84.535
Row softmax / NumPy 256 × 64 · f64	101836	101.836
Create zeros / Kernix 64 × 64 · f32	145	0.145

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	381	0.381
Create zeros / Candle 64 × 64 · f32	152	0.152
Create zeros / ndarray 64 × 64 · f32	90	0.09
Create zeros / faer 64 × 64 · f32	155	0.155
Create ones / faer 64 × 64 · f32	183	0.183
Create ones / NumPy 64 × 64 · f32	1339	1.339
Create ones / Kernix 64 × 64 · f32	183	0.183
Create ones / Candle 64 × 64 · f32	237	0.237
Create ones / ndarray 64 × 64 · f32	153	0.153